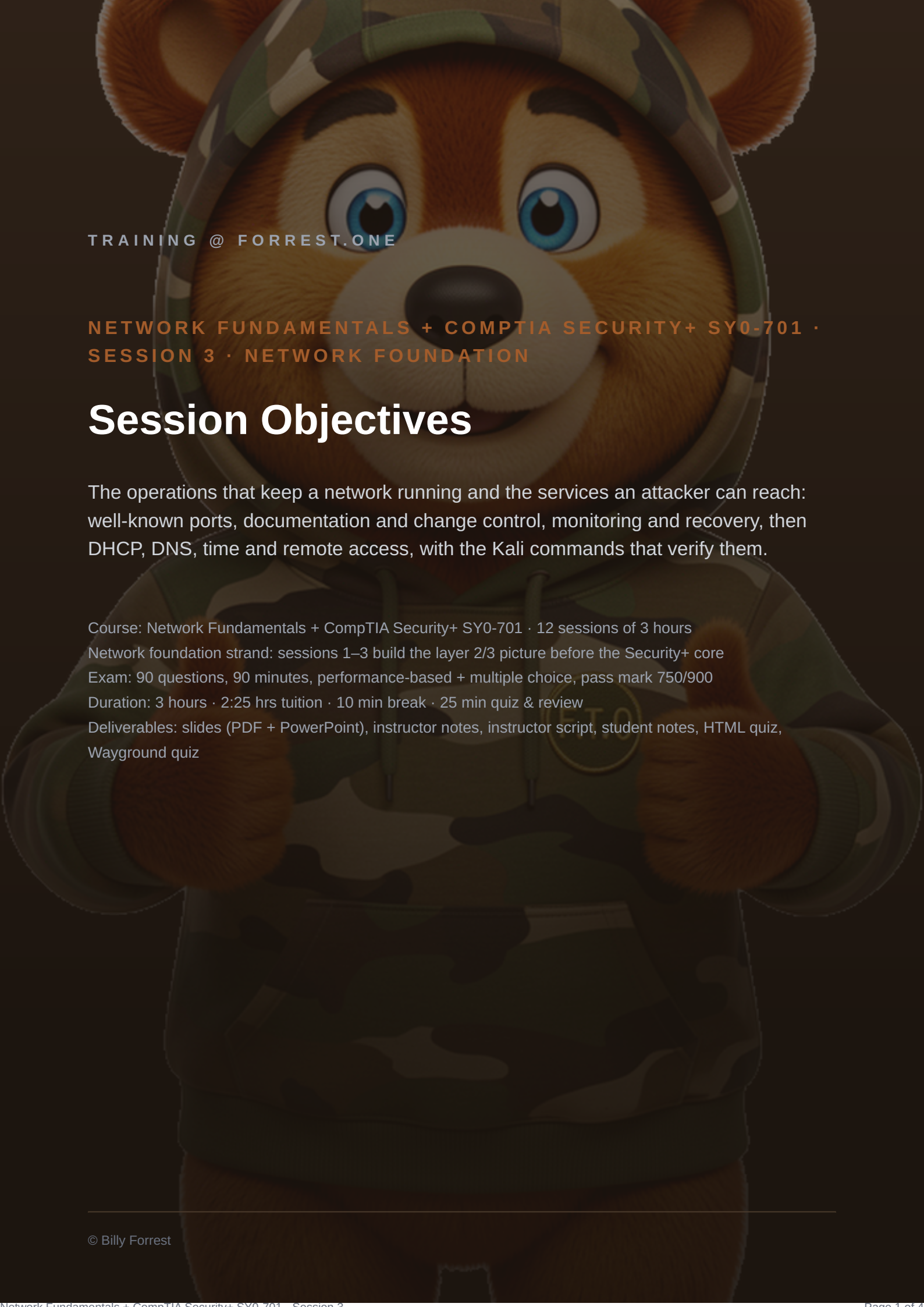




TRAINING @ FORREST.ONE

**NETWORK FUNDAMENTALS + COMPTIA SECURITY+ SY0-701 ·
SESSION 3 · NETWORK FOUNDATION**

Session Objectives

The operations that keep a network running and the services an attacker can reach: well-known ports, documentation and change control, monitoring and recovery, then DHCP, DNS, time and remote access, with the Kali commands that verify them.

Course: Network Fundamentals + CompTIA Security+ SY0-701 · 12 sessions of 3 hours

Network foundation strand: sessions 1–3 build the layer 2/3 picture before the Security+ core

Exam: 90 questions, 90 minutes, performance-based + multiple choice, pass mark 750/900

Duration: 3 hours · 2:25 hrs tuition · 10 min break · 25 min quiz & review

Deliverables: slides (PDF + PowerPoint), instructor notes, instructor script, student notes, HTML quiz, Wayground quiz

Session Objectives: Network Services & the Attack Surface

This third session closes the network-foundation strand. It covers the operations that keep a network honest (ports and protocols, documentation, change control, monitoring and recovery) and the services that make an IP network usable (DHCP, DNS, time synchronisation and remote access), with Kali Linux commands for verification. Each service is treated as an attack surface: a set of entry points an attacker can reach by a specific threat vector. Sessions 4 to 12 reference this material constantly, especially threats (Session 5), network security (Session 8) and security operations (Session 11).

Learning outcomes

By the end of the 3-hour session, candidates will be able to:

- Name the well-known ports for file transfer, remote access, email, web, name and management services.
- Distinguish plaintext services from their encrypted replacements, such as FTP against SFTP and Telnet against SSH.
- Describe unicast, broadcast, multicast and anycast traffic, and where each is used.
- Explain the difference between a physical and a logical network diagram, and the purpose of rack diagrams, cable maps, wiring schematics and floor plans.
- Explain asset inventory, IP address management and conflict detection.
- Interpret a service level agreement: uptime, response time, resolution time and the effect of each additional nine.
- Describe predictive and validation wireless site surveys, heat maps and non-overlapping channels.
- Explain End of Life against End of Support, and the update path with a tested rollback.
- Describe secure decommissioning: sanitisation, certificate and credential removal, destruction, degaussing and certified recycling.
- Explain change control: the Change Advisory Board, standard, normal and emergency change, and the contents of a change request.
- Explain configuration baselines, drift, compliance checking and configuration backup strategy.
- Describe SNMP manager, agent, object identifiers, GET, SET and trap, and the community string weakness.
- Compare flow data (NetFlow, sFlow, IPFIX) with full packet capture, and say when each is appropriate.
- Explain baselines, threshold alerting, anomaly detection and trending.
- Explain log aggregation, syslog severity levels, SIEM correlation and automation.
- Define RPO, RTO, MTTR and MTBF, and say what each drives.
- Compare cold, warm and hot recovery sites, and describe active-active, active-passive and load-balanced availability.

- Describe disaster recovery testing: tabletop, simulation, parallel test and full failover.
- Explain DHCP lease negotiation, relay agents, reservations, scope options and failover.
- Compare the IPv6 addressing methods: stateless autoconfiguration, stateful and stateless DHCPv6, and privacy extensions.
- Explain DNS hierarchy, record types, time to live, split horizon and DNSSEC.
- Explain NTP strata, the need for several sources, and what clock drift breaks.
- Describe VPN types, IPsec transport and tunnel modes, and full against split tunnelling.
- Choose between a console cable, SSH, a web interface and a programmatic interface for device management.
- Use dig, ntpq, ss and journalctl to verify services and read evidence on Kali Linux.

Course context

Item	Detail
Course	Network Fundamentals + CompTIA Security+ SY0-701, 12 sessions of 3 hours
Strand	Network foundation (sessions 1 to 3): the final session of the strand
Prerequisites	Sessions 1 and 2: models, addressing, switching, routing and architectures
Assessment	HTML learning quiz (20 questions) + Wayground quiz (15 questions)
Exam link	Ports, protocols and operations map to Security+ Domains 3 and 4

Session 3 coverage

Chapter	Area	What is covered
1	Ports, protocols and traffic types	Well-known ports and their encrypted twins; unicast, broadcast, multicast and anycast
2	Documentation, agreements and surveys	Physical and logical diagrams, specialised drawings, inventory and IPAM, service levels, wireless surveys
3	Life-cycle and controlled change	End of Life and End of Support, updates and rollback, secure disposal, change control
4	Configuration and monitoring tools	Baselines and drift, configuration backup, SNMP, flow data, packet capture
5	Baselines, logs and resilience	Baselines and metrics, log aggregation and SIEM, RPO and RTO, recovery sites, high availability, testing
6	Core services and remote access	DHCP, IPv6 addressing, DNS, time synchronisation, VPNs and device management

How this session maps to the course

Session 5 (threats) names DNS poisoning, on-path attacks and rogue devices as real attack vectors, all of which need the port knowledge from Chapter 1. Session 8 (network security) builds firewalls and VPNs on the port and tunnel material. Session 11 (security operations) uses the log, baseline and flow commands demonstrated here. Every quiz question is tagged with its topic area (e.g. 3A DNS).